

LECTURE-11

MATHEMATICS 1 (15B11MA111)

Lagrange Method for Undetermined Multipliers

Partial Differentiation CO[C105.2]

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 - R.K. Jain and S.R.K. Iyenger, "Advanced Engineering Mathematics" fifth edition, Narosa publishing house, 2016.

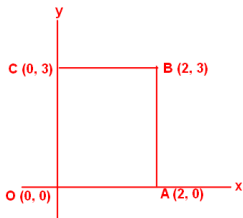
Numerical Problem

Find the absolute minima and absolute maxima of the function

$$f(x, y) = 4x^2 + 9y^2 - 8x - 12y + 4$$

over the rectangle in the first quadrant bounded by the lines $x = 2$ and $y = 3$ and the coordinate axes.

Solution: Since the function may attain maximum/minimum at the critical points or on the boundary of the rectangle shown in figure.



Cont...

We have

$$f_x = 8x - 8 = 0 \Rightarrow x = 1.$$

$$f_y = 18y - 12 = 0 \Rightarrow y = 2/3.$$

Hence $(1, 2/3)$ is the critical point. We find that

$$r = f_{xx} = 8, \quad s = f_{xy} = 0, \quad t = f_{yy} = 18.$$

$$rt - s^2 = 144.$$

At the point $(1, 2/3)$, we have

$$rt - s^2 = 144 > 0, \quad r = 8 > 0,$$

therefore the point $(1, 2/3)$, is a point of relative minima and the minimum value is $f(1, 2/3) = -4$.

Cont...

On the boundary line OA, we have $y = 0$ and

$$f(x, y) = f(x, 0) = g(x) = 4x^2 - 8x + 4,$$

which is a function of single variable x . For minima and maxima setting

$$\frac{dg}{dx} = 0 \Rightarrow 8x - 8 = 0 \Rightarrow x = 1.$$

Now

$$\frac{d^2g}{dx^2} = 8 > 0.$$

Therefore, at $x = 1$ the function has minimum and minimum value is $g(1) = f(1, 0) = 0$.

Also, at the corners $(0, 0)$ and $(2, 0)$, we have

$$f(0, 0) = g(0) = 4, \quad f(2, 0) = g(2) = 4.$$

Cont...

Along AB, we have $x = 2$ and

$$f(x, y) = f(2, y) = h(y) = 9y^2 - 12y + 4,$$

which is a function of single variable y . Now

$$\frac{dh}{dy} = 0 \Rightarrow 18y - 12 = 0 \Rightarrow y = 2/3.$$

Now

$$\frac{d^2h}{dy^2} = 18 > 0.$$

Therefore $y = 2/3$ is a point of minimum and minimum value is $h(2/3) = f(2, 2/3) = 0$. At corner $(2, 3)$ we have

$$f(2, 3) = h(3) = 49.$$

Cont...

Along BC, we have $y = 3$ and

$$f(x, y) = f(x, 3) = g(x) = 4x^2 - 8x + 49,$$

$$\frac{dg}{dx} = 0 \Rightarrow 8x - 8 = 0 \Rightarrow x = 1.$$

Now

$$\frac{d^2g}{dx^2} = 8 > 0.$$

Therefore at $x = 1$, the function has minimum and minimum value is $g(1) = f(1, 3) = 45$. Also, at the corner $(0, 3)$, we have $f(0, 3) = g(3) = 49$.

Cont...

Along CO, we have $x = 0$ and

$$h(y) = 9y^2 - 12y + 4,$$

which is same case as for $x = 2$.

Therefore, the absolute minimum value is -4 which occurs at $(1, 2/3)$ and the absolute maximum value is 49 which occurs at $(2, 3)$ and $(0, 3)$.

Conditional maximum/minimum

In some practical problems, we need to find the maximum/minimum value of a function $f(x_1, x_2, \dots, x_n)$ in which the variables are not independent but are connected by one or more constraints of the form

$$\phi(x_1, x_2, \dots, x_n) = 0, \quad i = 1, 2, 3, \dots, k,$$

where generally $n > k$.

Lagrange's Method of Undetermined Multipliers

To find the extremum of the function $f(x_1, x_2, \dots, x_n)$ with the condition

$$\phi(x_1, x_2, \dots, x_n) = 0, \quad i = 1, 2, 3, \dots, k. \quad (1)$$

We construct an auxiliary function of the form

$$F(x_1, x_2, \dots, x_n, \lambda_1, \lambda_2, \dots, \lambda_k) = f(x_1, x_2, \dots, x_n) + \sum_{i=1}^k \lambda_i \phi_i(x_1, x_2, \dots, x_n),$$

where λ_i are known parameters and are known as Lagrange multipliers.

Necessary Condition for critical points

The necessary condition to find the critical points are

$$\frac{\partial F}{\partial x_1} = 0 = \frac{\partial F}{\partial x_2} = \dots = \frac{\partial F}{\partial x_n} = 0,$$

which converts to

$$\frac{\partial f}{\partial x_j} + \sum_{i=1}^k \lambda_i \frac{\partial \phi_i}{\partial x_j} = 0, \quad j = 1, 2, \dots, n. \quad (2)$$

From equation (1) and (2) we get $(n + k)$ equations in $(n + k)$ unknowns $x_1, x_2, \dots, x_n, \lambda_1, \lambda_2, \dots, \lambda_k$. Solving these equations, we obtain the required critical points $(x_1, x_2, \dots, x_n)$ at which the function f has an extremum. Further investigation is needed to determine the exact nature of extreme points.

Numerical Problems

Example: Find the minimum value of the function

$$f(x, y) = x^2 + y^2 + z^2.,$$

subject to the condition

$$xyz = a^3.$$

Solution: Here $\phi(x, y, z) = xyz - a^3 = 0$. Now consider the auxiliary function

$$F(x, y, z, \lambda) = x^2 + y^2 + z^2 + \lambda(xyz - a^3).$$

We obtain the necessary condition for extreme points as

$$\frac{\partial F}{\partial x} = 2x + \lambda yz = 0, \quad \frac{\partial F}{\partial y} = 2y + \lambda xz = 0, \quad \frac{\partial F}{\partial z} = 2z + \lambda xy = 0.$$

Cont...

From these equations, we obtain

$$\lambda yz = -2x \Rightarrow \lambda xyz = -2x^2,$$

$$\lambda xz = -2y \Rightarrow \lambda xyz = -2y^2,$$

$$\lambda xy = -2z \Rightarrow \lambda xyz = -2z^2,$$

therefore, we get $x^2 = y^2 = z^2$. Using the condition $xyz = a^3$, we get the extreme points as

$$(a, a, a), (a, -a, -a), (-a, a, -a), (-a, -a, a).$$

At all these points the value of the given function is

$$x^2 + y^2 + z^2 = 3a^2.$$

Cont...

Also, the arithmetic mean (*AM*) of x^2 , y^2 , z^2 is

$$(x^2 + y^2 + z^2)/3$$

and geometric mean (*GM*) is

$$(x^2 y^2 z^2)^{1/3} = a^2.$$

Since

$$AM \geq GM,$$

therefore we get

$$x^2 + y^2 + z^2 \geq 3a^2.$$

Hence all the above extreme point are constraint/conditional minimum and the minimum value of $x^2 + y^2 + z^2$ is $3a^2$.

Cont...

Example: Find the extreme values of the function

$$f(x, y) = 2x + 3y + z,$$

subject to the conditions $x^2 + y^2 = 5$ and $x + z = 1$.

Solution: Here $\phi_1(x, y, z) = x^2 + y^2 - 5 = 0$ and $\phi_2(x, y, z) = x + z - 1 = 0$. Now consider the auxiliary function

$$F(x, y, z, \lambda_1, \lambda_2) = 2x + 3y + z + \lambda_1(x^2 + y^2 - 5) + \lambda_2(x + z - 1).$$

We obtain the necessary condition for extremum as

$$\frac{\partial F}{\partial x} = 2 + 2\lambda_1 x + \lambda_2 = 0, \quad \frac{\partial F}{\partial y} = 3 + 2\lambda_1 y = 0, \quad \frac{\partial F}{\partial z} = 1 + \lambda_2 = 0.$$

Cont...

On solving these equation, we get $\lambda_1 = \pm 1/\sqrt{2}$, $\lambda_1 = -1$ and the extreme points as

$$(-1/\sqrt{2}, 3/\sqrt{2}, 1 + 1/\sqrt{2}), (1/\sqrt{2}, 3/\sqrt{2}, 1 - 1/\sqrt{2})$$

also the extreme values of the function $2x + 3y + z$ at these two points are $1 - 5\sqrt{2}$ and $1 + 5\sqrt{2}$ respectively.

Example: Find the minimum value of the function

$$f(x, y) = x^2 + y^2 + z^2,$$

subject to the condition $ax + by + cz = p$.

Solution: Do yourself.

Thank
you !